



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.7

Generate Equivalent Expressions

Lesson 6-8 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can use the commutative, associative, and identity properties to rewrite expressions.

Language Objective: I can explain my reasoning using the words commutative property, associative property, and identity property.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Properties of Operations

Commutative Property

Associative Property

Identity Property

Property

Distributive Property



Tap any word to see its meaning and a picture.

1

Rules that let you reorder or regroup without changing the

2

The rule that order does not change a sum or product, like $a + b = b + a$, is the

3

The rule that grouping does not change a sum or product, like $(a + b) + c = a + (b + c)$, is the

4

The rule that adding 0 or multiplying by 1 keeps a number the same is the

5

A rule that is always true in math is a

- 6 The rule that lets you multiply a number across a sum, like $a(b + c) = a \times b + a \times c$, is the .

Write About the Math

What properties did the engineer use, and why does rearranging make the calculation easier?

¿Qué propiedades usó la ingeniera, y por qué reacomodar hace más fácil el cálculo?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Properties of Operations**

Propiedades de las operaciones

☐ **Commutative Property**

Propiedad conmutativa

☐ **Associative Property**

Propiedad asociativa

☐ **Identity Property**

Propiedad de identidad

☐ **Property**

Propiedad

Model: *The commutative property lets you change the order of a sum without changing the total. — Students name the commutative property and explain that $3 + 5 + 2$ gives 10 in any order because reordering addends does not change the sum.*

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The engineer used the ____ property to change the ____ and the ____ property to change the _____. This makes it easier because $4 \times 25 =$ ____, which is a friendly number to multiply by.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- My evidence is ____, so ____.

Mi afirmación es ____.

Mi evidencia es _____, entonces _____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Properties of Operations
2. **Notes 2:** Commutative Property
3. **Notes 3:** Associative Property
4. **Notes 4:** Identity Property
5. **Notes 5:** Property
6. **Notes 6:** Distributive Property
7. **Watch for this mistake:** *A common mistake in Properties of Operations is applying the commutative property where it does not belong, or confusing it with the associative property. Students often assume $15 - 8 = 8 - 15$ (it does not: $7 \neq -7$ — commutative only works for addition and multiplication), or they see parentheses move and call it "associative" even when the numbers inside just swapped places, like $(2 + 3) + 4 = (3 + 2) + 4$ (that is commutative — the order inside the parentheses changed, not the grouping). Before you label a property, ask: did only the order change (commutative), did only the grouping change (associative), or was a 0 or 1 involved (identity)?*
8. **Try It 1:** C. Associative Property — *The grouping (parentheses) changed but the order of the numbers stayed the same, so this is the Associative Property.*
9. **Try It 2:** D. Identity Property — *Multiplying by 1 keeps the same value, so this is the Identity Property of Multiplication.*